CHAPTER TWO

LITERATURE REVIEW

2.1 Advancements in Data-Driven Prediction Techniques

In order to design the successful programs in the context of preventative care, the malaria burden is projected to be sufficiently large, that is, in sub-Saharan Africa, where malaria has developed into a horrible economic and health challenge to the families at risk (Kamau et al., 2020). Average and total data utilized by history which is in general use in the traditional methods does not capture individual variation in the malaria rates among the states and societies. It is owing to that fact that such strategies may give a fictitious account on the true burden of malaria and render implementation efforts less successful.

Due to the shortcomings of traditional approaches, we see the current trend of the sophisticated state-of-the-art data-based approaches that would provide more precise and detailed projections of malaria. There are numerous considerations when compared to the general data utilized in the old methods, which only intends to account as much factors as possible via the environment, population specifications, a cost of treatment and location trends with the aim of reaching a more appropriate and accurate prediction.

These sophisticated models are able to assist in perfecting local hotspots, seasonal change and even suit together rainfall, temperature and social economic level in a single analysis. The current strategies of malaria risk mapping have been applicable in malaria risk mapping using malaria risk maps or Bayesian spatial models, and also in estimating the prevalence of the amount of disease that exists using hierarchical regression. As indicatively revealed by Lindtjorn et al. (2017), the

combination of factors between economic data and location data also leads to the existence of inconsistency in how individuals who reside in Ethiopia spend on malaria control and prevention. Kamau et al. (2020) also discovered that the Pakistan and place information included in the prediction of the malaria also helped the prediction in Kenya.

This is merely one of the indications that it is of great significance to have models that can process large and complex data, and is not difficult to decipher by policy makers and by the health planners.

2.2 Introduction to Predictive Systems in Public Health

Evidence-based information has triggered the need of the predictive systems in the field of citizen health and this can assist the government and any other organization in responding on time and allocate the available resources in a most effective way achievable. The current research by Kumar et al. (2016, 2017) demonstrates the fact that machine learning algorithms (decision trees, neural networks and support vectors machines) could be applied to enhance the detection and risk assessment of malaria. These applications operate on comprehensive surveillance and lead to the scenario where the health planners determine the nature of the infection process and predicts the outbreaks more efficiently as compared to the traditional ones. The strategy transforms raw data into the information to work with, and helps to allocate the preventive means in a certain manner, i.e. purchasing target distribution of preventive tools, i.e. insecticide-treated bed nets and antimalarial medication (Kumar et al., 2016; Kumar et al., 2017).

The principal predictors of malaria are spatial temporal analysis and climate variables, as it has been demonstrated in other studies by Khatua et al. (2019) and Parvin et al. (2019). Examples of the environmental conditions that are used by such

models include the temperature, the rain that falls and the humidity required in the breeding of the mosquito. In this regard, the systems can generate fine risk maps of areas that are quite likely to be affected by malaria. These real-time maps put the health authorities in control due to Geographic Information Systems (GIS) and remote sensing systems. They assist in the timely measure of spraying the indoors residences and select health target area works. The plan will increase the resource allocation process, and the controls improvement (Khatua et al., 2019; Parvin et al., 2019).

With all these developments, however, there are still issues. The predictive models must be viable and easy to understand by the local health officials. The models developed on the information about one area will not work effectively in other areas because of the differences in health facilities and the comparative socio- economical conditions. This shows that it needs to display non rigid approach to it including formal additive regression models. These models have the ability of addressing the linear, non-linear and spatial effects to model the complexity in the malaria transmission in different environments. Such powerful technologies would allow the administration and controls of malaria and resources allocation at the community most inclined to this disease by correlating the technical prediction with highly practical interventions of the state of the public health. All the aforementioned authors have been mentioned in this discussion.

2.3 Prediction of Malaria Burden: A Worldwide Perspective

Malaria has been a major concern of the humanity and even more so in the sub-Saharan African region of the globe, as well as in parts of Asia and South America where legions of the populace continue to face the risk of being malarial attacked and dying. The relevant disposition towards the malaria prevention planning is also

now dictorial in the process of mediating specific intervention, strengthening resource allocation and propagating control mechanisms. Because of the predictive model, which allows the use of climate information, geographical variation, demographic variation, and access to treatment, the malaria predictive abilities have been enhanced.

Among the greatest accomplishments is artificial prediction of the malaria epidemics in most of the countries that have started utilizing sophisticated statistical and mathematical tools to forecast its economic effects. Giving an example, Kamau et al. (2020) investigated how climate variability and temperature and rainfall relate with malaria prevalence visibility in Kenyon through the spatial and spatial temporal regression analysis. In the same manner, Lindtjorn et al. (2017) have observed the economic cost of malaria on the Ethiopian family and they have discovered the usefulness of the models that enable the spatial deviation and household risks to be examined.

2.3.1 Case Studies of Malaria Burden Forecasting in Different Countries

The African nations that have option, as last couple of years have demonstrated, have started to explore more intricate models to appreciate the effects of malaria on the people, and vice versa to present the variations to the manufacturing of some interventions. Simultaneously, data indicated in the domestic survey based on the Ethiopian population to estimate the economic price of malaria was provided to Tefera (2021), and the latter were released without an opportunity to consider the specifics of the space. The possibilities of the diseases like the spreading of malaria in Kenya in the article by Kamau et al. (2020) were better predicted in their article that used the weather modes in their models like rain, temperature, and the latter is known to have a potential in the hotspots of diseases. The authors of the Noor et al.

(2008) article applied the Bayesian geostatistical models to the Tanzania and came up with some details about the malaria risk maps which could aid in inoculating more bed nets and indoor sprays. Malaria management by the states of the Southeast Asians facilitates integration of the environmental intelligences and real time reflection of the malaria management by putting into practice the practices involved. These pictures are appealing to more feminine structures like STAR model which can bear the expenses in the house like seasonal dynamics in the house to bring malaria prediction to action.

2.3.2 Malaria situation in Nigeria

Malaria crisis has stayed to be experienced in Nigeria whereby by the reported malaria cases it is estimated that a quarter and nearly three times of the overall death in Nigeria is caused by malaria (WHO, 2023). Nigeria is the luckiest state in the sub-Saharan region since the state is the largest donor, and has been engaging in the giant investment in the preventative measure such as provision of the insecticide treated net, household spraying in addition to the provision of the diagnosis/ treatment services. Nevertheless, the malaria has been one of the most multipurpose conditions that have caused morbidity and mortality.

Distribution of malaria is not even in Nigeria. More recent evidence indicates 2021 Malaria Indicator Survey of prevalence rate of the parasites among children below the age of five years in Nigeria which showed that the national prevalence was 22 percent and prevalence of 34 and significantly lower prevalence at 34 in both the north and southwest respectively. It is also found in the aspects of the nature of transmission which assume the shape of unavailability of health care centers, resistance to insecticides, and social-economic chaos.

The two categories of low-income families and rural families are likely to be one of the least affluent afflicted by the condition since they have to pay very high out of pocket prices in an effort to receive treatment to prevent the disease. It can swallow the considerable share of the family savings and become even worse when it comes to direct costs, i.e. medications, doctor visits, traveling, etc. can also be added (Affiah et al., 2022). Onwujekwe et al. (2013) also argued that loss of working days and productivity in terms of indirect expenses induce cumbersome burdens in the family, at a daily level. It has health related consequences of malaria but on the other hand, the cycle also leads to poverty due to the fact that the money used to purchase other basic necessities such as education and food is instead spent on malaria.

The survey that is undertaken on a countrywide basis, is not comprehensive enough to establish the hotspots or the malaria burden of the impact it has on the households with most of the situations. Okeke et al. (2021) emphasized the fact that a heterogeneously distributed malaria risk should be developed since to accomplish it, more specific data is essential in the Niger Delta. It sheds light on the significance of high predictive instruments that will collate the information of surveillance to space, economic and environmental mean and capability to allocate increased assets to the malaria battle in Nigeria, and evenly scatter it.

2.4 Literature review overview

The element of the real time forecasting of the weight of the malaria illness is an intriguing one in modeling the individualistic effective response of the economy of the individuals to the malaria minstrel specifically in the sub Saharan Africa where the malaria mass to the individuals has been colossal amid the susceptible households in economic and medical domains. The historical method of calculating the problem, using the historical records and overall average has not been able to

reach far in its attempt to project the frequent malaria outbreaks in the areas and other people. In the context of this, the effect of the malaria can be moderated using such mechanisms and the effect created under some intervention can be constrained.

In this respect, in this sense, the greater part of the dynamic models such as semi parametric Structured Additive Regression (STAR) model has been uploaded as the statistical modeling has emerged. This variation can also be parallelogated where the socioeconomic saves, access to the treatment would be factored in the options of the economic costs available to the STAR model in such a way that the mechanism would modify the variation in the interaction between local and seasonal variations of malaria risk other than the forecast available at the homes.

The answer to the utility of the STAR model is that, the answer is the opposite of more complex tools, which is not as straight forward to an extent such that the data will be presented in a way that will allow the policy makers within the health sector to make decisions. The reason is that it is able to consider space and time difference which is household level information that is the one that is desired such that the space that economy was lacking is occurring and the reaction against the malaria enlightened. The latter part will entail the retrospective that will examine the past researches on the malaria burdens prediction like the one by Tefera (2021) and the likelihood of whether the STAR model will be capable of enhancing the validity, the transparency and the policy applicability of the prediction tools of the high-burden areas.

2.5 Malaria household prediction predictors.

Malaria problem in a given household is a mystery that can be demystified with the help of different variables to consider and put together to predict the effects of the

problem sufficiently. They are the social and demographic background, status, and the social-economic situation in general.

Socio-Economic Factors

The socioeconomic are very crucial in determining the response of the households to the malaria and how to tackle the malaria problems. The amount that is incurred on prevention is the amount as is usually determined by the circumstance of the family budget of the income. The better income will also help deal with timely diagnosis and treatment and the inability to afford the money can make the family postpone an appointment with a doctor or even get less efficient and inexpensive treatment. The second reason is literacy: the failure to read or education of the house provider can be a barrier of comprehending the information about health, awareness of the symptoms, or undergoing the preventative procedure. Another important factor is the nature of the work of the main earner; those who earn low salaries, not registered and workers outside are at a greater risk of the malaria and will probably not have enough money to meet the treatment cost. The larger families would suffer because they would be shielding and curing the multitude of the other individuals and would have fewer resources to repel them since they will have limited mosquito nets. The poor housing furnishings and housing facilities, which is characterized by easy accessibility of the walls, floors or roofs by the mosquito, poor sanitation also adds to stagnant water, which serves as a breeding ground to the breeding of the mosquito.

The other button would be the access to the safe drinking water. They are more prone to it, as the families are dependent on the unprotected provision, and the number of mosquitoes grows. The other ways in which the risk can be escalated is by cooking

the food in wood which adds more attractiveness to the mosquitoes. It is also relevant to the supply of electricity; we can refer to the fact that with the lack of it, the families will not be able to take the subsequent precautionary measure as insecticide-treated nets or repulsions or even get information on health in time and, therefore, they will delay the treatment. This information gap can be filled with other bad things. The younger member of the family that is below the age of five years is more burdensome as young children are easily infected by malaria such that, there is increased health care escalation and increased cost. The thing about these risks is the worst as the family is giving birth to child, the malaria that has been passed to both the pregnant mother and the child could in both instances be a fatal health condition that would inflict even more economic burden on the families that are already stressed. They are traceable to the attributable socioeconomic issues and have massive implication on the quality life and health security, cost-effective conditions, and overall happiness of the malarial-infested households. Such dynamics ought to be acquainted so as to enable appropriate actions of prediction and reaction to the impacts of malaria among various communities.

Economic Factors

The family has paid money as a result of this malaria and this therefore can be mentioned to be direct costs and indirect costs and as it is, it is paramount to say that with regards to malaria-management a specific family has expended money differently.

Direct Medical

Some of the contributions are also direct medical costs to families which must cope with malaria. These expenses cover the whole contents required in the treatment

such as the price of purchasing anti malarial medicine, the price of meeting a doctor, the price of carrying out diagnostic tests. The families in most of the low income regions tend to finance such medical services through their own cash since they cannot gain access to insurance benefits such as government subsidy. The Malaria control is reported to generate revenue through the purchase of artemininin containing mixtures of drugs (ACTs), pain killers, and in some cases, antibiotics by the families residing in malaria prone areas (Sicuri et al., 2013). The disparity in the general expenses may differ significantly based on the purchasing ability of the citizens regarding branded or generic drugs and their capacity to cover the confirmed treatment. At the overwhelming number of rural places, the high rates of the lack of drugs in the community hospitals constantly compel the families to address the services of the private pharmacies where the prices are more expensive (Onwujekwe et al., 2013).

Then there is the other direct cost of diagnostic tests not including medication. A combination of microscopy and rapid diagnostic tests (RDTs) should be used in the diagnosis of malaria, as in this manner the unnecessary consumption of antimalarial drugs will be avoided; however, it will be increased further to the budget. According to White et al. (2011), the laboratory confirmation forms a big portion of the total treatment expenses. The poorer families; particularly the remote families can afford not to take the confirmatory tests but turn directly to the symptomatic treatment based on suspicion that is sure to become wasted on the development of resistance in the drugs. According to Deressa et al. (2007) as well, the expense of attending the hospital or even a private hospital and the off hand payments which have to be attended earlier, compounds the financial strains of the households. Its very expense might deter an immediate treatment on low income earners and lead to poor health results and consequent rise in the future health care expenditures.

Direct Non-Medical Cost

The treatment costs that might not be intertwined with medical treatments, in this case, the malaria disease, might be expensive and might be ignored by the families in estimating the costs. These are associated with the cost of the transportation in and out of the health centers, lunch during the treatment trips, accommodation, in case of an overnight stay. According to Sicuri et al. (2013), the amount spent on transport can consume a huge percentage of the household spending particularly in the rural families where people may need to cover long distances to access its services. In other instances, Wiseman et al. (1995) realized the cost of transport was removed to the later position immediately after spending on medicines in the household budgets. The local healthcare centers may not be sufficient enough and thus the families will be compelled to travel to the local healthcare centers and this adds to the increased financial burden as the family will have to use motorcycles, cars or boats.

Home based care is also expensive to many families even before they can consider seeking formal medical assistance. It can include purchasing of over the counter medicine, alternative medicine or visiting the services of healers in the place to cure the symptoms such as fever or possible malaria (Onwujekwe et al., 2013). It is due to the fact that these options appear to be less costly or cheaper compared to official healthcare and that is why families resort to such options. Nevertheless, assuming that these decisions will postpone the adequate treatment, aggravate the situation with the disease, and be accompanied with even higher costs as a follow-up is possible. According to Deressa et al. (2007) such drawbacks have led to increased financial burdens and put households at a risk of vulnerability especially when malaria seasons are at their highest. Non-medical costs greatly contribute to the

overall cost of the malaria episode as they may cost the household savings and hardly can create a situation where the family is forced to either borrow to cover their medical expenses or sell their assets.

Indirect Cost

The indirect costs which are the time wastage at the workplace, reduction of income or reduced output at the malaria is a long term effect to the family which is more likely to be encountered as compared to the direct costs. When any of the house members is victimized to malaria, another member of the family, who is typically a male, is compelled to miss work or quit farming and therefore lose his or her income. This is a nuisance that may be extremely challenging in the social set up of the poor rural families. Sicuri et al. (2013) also prove that these opportunity costs can be more significant than direct therapy costs.

The small scale farmers, daily wage earners and informal workers are the vulnerable groups that are particularly susceptible to the belt since their means of livelihood depends on their daily work. The malaria can result in the harboring of delayed farm activities as it reduces the crop production through the process of disrupting the planting or harvesting of crops through the largest farming seasons (White et al., 2011). The virtue that the malaria has taken a long period further complicates the issue and this makes the economical issue difficult. The malaria indicators of castro et al. (2004) revealed that the highest percentage of malaria burden in Mozambique is made up of the time lost in the malaria disease and prevention measures. The children are also deprived a chance to attend school through malaria and this may affect their future earning power. The families have languished in the poverty trap of low productivity where it is hard to save or invest in good houses, food and

education due to the ever-present malaria illness as per Deressa et al. (2007). This endless low-cost burden can propel the families to poverty thus even harder to emerge out of the conditions that predetermines people to malaria.

Socio-Demographic Factors

The coping of Malaria families will largely depend on the age of the head of the household. The young heads of household might be immature and less resource endowed to make good health decisions that would limit their potentials of accessing good prevention and treatment. Indeed, the elder heads would have a more realistic experience, and a more supportive circle, but they may be physically impaired in the ability to care of ill members of the family. It is also fuelled by weakness of some of the members of the household such as children below the age of five years and the old population. These groups are more likely to develop serious complications with the need to devote more time and medical attention and other care with the result of increased direct and indirect costs.

Healthcare and treatment are immensely affected under gender roles in a family set up. Women were heads of family and rural or poor areas are more likely to experience greater financial limitations and social barriers to access to transport, treatment or preventive equipment as bed nets. Traditions may deprive women of their entitlement to make their own health-related choices and delay the necessary treatment. It is also based on the education of the head of the house. The better educated people are inclined to be more aware of the ways of preventing malaria and associated symptoms and treatment that are bound to make the disease easier to identify and treat. Conversely, lack of education might create a false impression on

malaria and, thus, the families will not be willing to prevent malaria or to seek timely treatment.

The occupation of a head of household affects the income, as well as the risk of malaria. The people who are exposed to the open air or agricultural sector are more vulnerable to the mosquito bites and the people in unreliable and poorly paid jobs are less equipped with health care. The awareness of the fact that the head of the household was working last week will give a clue of how that family is doing in regards to their economic position. This may be because of the regular lack of sickness that would reduce incomes, use more poverty and malaria recovery. The effect of malaria on households is resultantly a combination of such variables as age, gender, education, occupation, and work stability. These factors are well understood in order to develop a good method of predicting and reducing the impact of malaria.

2.6 Old techniques of Malaria Burden prediction.

Most of the conventionally implemented predictive health outcomes on the malaria disease burden have incorporated the combination of regular health data, post-factum household survey, and basic descriptive statistics (Gething et al., 2011). Among the common tricks are the average incidences and trend analysis and simple correlation of the cases of malaria to such environmental factors like rain and temperature. Though they give a generalized picture of a national or even a regional picture, they are less likely to miss out on the minute details which are enclosed in a given community or family. Having determined that mapping of the malaria through fewer surveys carried out through routine health registries at the onset of the exercise gave generalized analysis of the risks, which failed to furnish the local difference of the disease transmission or the difference in the other continuously changing socio-economic landscape, Gething et al. (2011) did not achieve the intended effect.

The article by Hailu et al. (2017) is one of the examples of such articles, and it concerns cost of malaria in south-central Ethiopia and it presents a household survey that contained details about direct cost of treatment and loss of lost work income. It was a compilation that was founded on a structured interview and portrayed a more fascinating picture on the economic cost of malaria at family level. Nonetheless, geographical differences, climatic and the environment is one of the factors, which their model failed to take into account and which might affect the propagation of malaria. Similarly, the study by Okello et al. (2006) on the prevalence of malaria in the Ugandan highland employed archival data on the health facilities and crude environmental data on the rain and temperature but their estimation failed to put into consideration more elegant geographic models and neighborhood environmental and socio-economic conditions which are likely to contribute to the household risk. All these descriptions make one think that although the standard models illustrate the overall trends of the malaria burden, it is in most cases not the data that is applied at a more precise and an operational level that would be required by the policymakers and the heads of the health sector.

These weaknesses especially have resulted in the formation of improved and multi-dimensional models that may be worth considering in the malaria forecasting field that incorporate other elements. Admittedly, there is a massive advantage to semi-parametric or structured additive model Structured Additive Regression (STAR) model. The space, time and environment data combined with the traditional survey data on health will enhance and curb the more precise risk estimates that are situation-specific. To the extent that these progressive methods will be able to retrieve the impact of the changing climatic patterns, land utilization, population migration, family socioeconomic status and malaria burden at local levels as opposed to the direct averages and fixed analysis. The future development of the

intervention under consideration involves the enhancement of this modeling because the allocation of resources and the level of effort invested in the malaria reduction should be based on the curves and bends in the malaria patterns of transmission, namely in such a country as Nigeria, where the malaria patterns of transmission do not appear to be homogenous.

2.7 Malaria burden prediction Statistical Semi-parametric Model types.

Other more modern tendencies in the prediction of the burden of malaria are such that a change of attitude is that parametric statistical models have been replaced by more flexible semi-parametric models that is a trade-off of understanding versus flexibility. The optimal model that is supposed to be employed to represent such change is the Generalized Additive Model (GAM) due to the fact that the extension of the linear models is the use of smooth and non-linear functions. The GAMs are useful when environment and the social factors are complicated and work towards the propagation of the malaria disease. Sahu et al. (2018) have also shown in their illustration that the GAMs could be useful in estimating the spatial and seasonal pattern of the disease risks. It needs such a flexibility in which the elements of rainfall, temperature and vegetation are not linear since they can not be expressed in the traditional models. This is caused by the fact that it is the efficacy of GAMs that subsequently allow the researcher to model local transmission loops that are more fruitful in a termination of more narrowly focused interventions. The other relevance tool is the Bayesian hierarchical model (BHM) that is a semi-parametric tool as well as the GAMs. Specifically, the BHMs can be used when it comes to spatial relationships and character hierarchy of malaria surveillance data. Diggle et al.

(2002) have established that BHMs are capable of executing the mapping of the instances of malaria because they exploit random effects as well in an attempt to generate the grid of observed and unobserved heterogeneity. BHams allows developing complex detailed and accurate risk maps using the variation across different levels e.g. individuals to communities by using incomplete or biased information. Their ability to combine the historic data they have with the knowledge of the geographical distribution allows them to make weaker estimations and their ability to allocate resources in the region of public health is more successful.

This is because GAMs and BHMs have been in a position to make enormous contributions towards prediction of malaria as they offer dynamic tools of analysis to researchers and health planners in other epidemiology areas. The models produce place-based predictions which are carefully used in the process of coming up with place-based interventions, efficient resource allocation and efficient early warning mechanisms. They give extra reality to the naked models in the quantification of the complex relationships and the space patterns of the malaria transmission. The need to apply the semi-parametric statistical methods, in turn, can be argued as a necessary measure to take in the malaria study because it enables estimating the health status of the population more precisely and, perhaps, introduce the intervention with even more beneficial consequences (Sahu et al., 2018; Diggle et al., 2002).

2.8 Semi Parametric Statistical Modelling Importance of Public Health.

The methodology of the public health analysis is associated with the application of statistical modeling in case of the infectious agent (malaria) in stake, and the mechanism is too complicated and dynamic (Kamau et al., 2020). Correct and precise models and predictions on the disease burden in the malaria prone areas must be carried out in a way that would facilitate the mobilization of interventions in good

time, planning and resource allocations. These conventional methods that tend to rely on the old experience or judgment of human are not very flexible in identifying changes that occur in real-time or local parameters (Hailu et al., 2017).

The semi-parametric are more significant because the author could include nonlinear, spatial, and even temporal relationships within the same regression considering that Structured Additive Regression (STAR) model is the most accommodative model. They can then be used in predicting the large and local effects of malaria (Kneib and Fahrmeir, 2007). In comparison to other more traditional models that use linear regression or machine learning exclusively, the STAR model allows considering the issue of malaria in a more significant context, and such aspects as weather patterns, household demographics, the cost of treatment, or geographic disparities are taken into consideration.

In practice, the implementation of STAR modeling may help the researchers and the health authorities overall to know the high-risk areas, estimate the direct and indirect costs, and create the interventions with greater confidence. Specifically, Lindtjorn et al. (2017) showed that economic and spatial data was vital to illustrate the inequality in the burden of malaria within the rural households in Ethiopia. Equally, Affiah et al. (2022) have established that even household expenses analysis on the practices of those seeking treatment and care in Akwa-Ibom, Nigeria can help establish some of the latent shortcomings. Stat models offer the highly desired accuracy and insight in the interpretation of data in order to implement the data in the right health programs. However, in contrast to deep learning or opaque models, which are not always interpretable, STAR models explain the impact of many factors on the final results and, thereby, help those representing communities and countries when making a choice (Sahu et al., 2018). It must also be borne in mind that, one must avail of organized statistical model (especially hyphenated model of STAR) to help us in the early decision making on malaria burden as we are able to conduct

multifactorial, time-based and spatial analysis. This is one of the ways, through which the raw data will be converted into the applicable knowledge, on how to plan more effectively, set up some policies and create modes of control of malaria in the sustainable manner.

2.9 Implementation of Technologies of Statistical semi-parametric Model in Nigeria.

The tradition of predictive analytics has taken the center stage in the health department of Nigeria especially in predicting the burden of malaria. The latest is directed towards implementing more sophisticated semi-parametric and computational procedures in a bid to increase the accuracy and impact. Indicatively, Oluwagbemi et al. (2012) designed a computational system of GIS taking into account the modeling of cases of malaria in Africa when interventions occur under different circumstances. Their findings have shown that by the year 2014 India should be able to reduce cases of malaria by 46.6 percent in case 90 percent more malaria control interventions are carried out. This observation implies that predictive systems are capable of transforming the malaria surveillance into a dynamic process instead of the static process thereby providing the policy makers with the much-needed information in resource allocation and intervention planning as well. Moreover, the analysis of 2010 Nigeria Malaria Indicator Survey that has demonstrated that there is a considerable geographical distribution of anaemia risk among children and that most of it is concentrated along the northern regions was done with the assistance of Bayesian semiparametric model (Gayawan et al., 2014). These studies indicate that there is need to take into account the spatial elaboration of the determination of the malaria burden so that specific steps can be taken.

In addition to the malaria specific studies, the advanced level of implementing the special statistical techniques in the health research in Nigeria is also being embraced. Indicatively, Fagbamigbe and others (2018) analysed using Demographic and Health Survey the postpartum abstinence periods and semi-parametric methods to do the survival analysis, which was the Kaplan-Meier and the Cox regression methods. This notwithstanding the fact that this study is not malaria specific as such, it is an overall trend because it entails the application of modern statistical indicators in analyzing the health behaviours of both a household and a region. Likewise, Dontsop et al. (2012) also established that Local Average Treatment Effect (LATE) was utilized to identify the effect of high-quality rice varieties on farm families in Nigeria based on productivity. The cost and effectiveness of malaria interventions can be analyzed using this type of rigour of methods as well.

With all these said and done, there are still a number of huddles in the process of enhancing the predictive malaria in Nigeria in terms of burdens estimation. The most challenging ones are unavailability of good geo-referencing health and environmental data, lack of computer facilities and inaccessibility of local personnel with expertise to develop and analyse advanced models. Gayawan et al. (2014) noted that the elaboration of the spectrum of successful spatial maps can play a crucial role in directing some interventions on the state-level towards the vulnerable children, and the resolutions can be built on the case of the malaria control campaigns. The result of bridging the gap between the national malaria estimates and the actual experiences that the families go through can be augmented through the establishment of the required technical capacity and connection of the semi-parametric models and geospatial to the household cost data. This would help in having more specific, informed and effective malaria control projects.

2.10 Related Literature

Other studies have gone beyond the analysis of malaria to examine the other aspects besides the direct cost analysis. These include:

Tefera et al. (2020)

In this paper, the impact of malaria on rural families in the Chewaka District in Western Ethiopia has been measured economically. Despite the reports rendered nationally, whereby cases of malaria have been minimized, the illness remains a looming danger to the health and economic sustainability of 60 percent of the locals. The paper has quantified both direct and indirect costs of malaria and also perceived socioeconomic determinants of the same burden. The research used the community-based cross-sectional study design and retrospective cost to carry out a household survey on 765 households. Indeed, the multivariate logistic regression analysis revealed that the prevalence rate was 32 and the mean of cases of malaria per household per year was two and the mean of cost per household USD approximately 16 with 78 percent of the breach on the indirect cost that was due to the result of lost working days. Several family households were forced to spend over five percent of their income on treating malaria. The sex and education level of the head of the household, mode of transport, incidence and duration of bouts of malaria, and modes of diagnosis were the major socioeconomic predictors. The latter was indicated as one of the major study limitations.

Affiah et al. (2022)

The paper assessed the economic consequences of treatment/health seeking behavior of malaria in the Akwa Ibom State, Nigeria where 97 percent of the residents are

susceptible to malaria and severe epidemics are a common occurrence. The research was conducted as a cross sectional survey among 640 households in three local governments and it found that 55.7 percent of the sampled claimed that they would choose the drug stores in matters related to treatment. The average direct and indirect cost about the households had to be 8,563.77 Naira (almost 22.60 USD) and 10,437.09 Naira (almost 27.54 USD), respectively, and the average was 18,868.10 Naira (almost 49.78 USD) a year. When it comes to the malaria disease, the low-income households spent as much of as 36 percent of their monthly revenues on treating the disease. This shows that malaria management is economically expensive to families and that is why more efforts should be directed into prevention and management. However, the study was analyzed descriptively but there were no predictive and spatial modeling that can be used to provide a better place household-level burden estimation.

Khan et al. (2024)

The purpose of this paper was to develop and compute machine learning models that are capable of predicting the malaria outbreak occurrence in the Gambia and address the long-standing problem of seasonal malaria outbreaks that are difficult to predict in other ways such as traditional surveillance. Several machine learning models as well as their output were compared in the study i.e. the random forest and the support vector machine and the logistic regression by using the historical records of malaria and climate i.e. the rainfall, temperature and humidity. The greatest predictive value was achieved with the assistance of the Random Forest model where the climate variables and the population density were determined as the material factors of an outbreak. The researchers discovered that machine learning has the potential of supporting the accuracy and real time of the malaria outbreak forecasts, although the greatest restriction was that it relied on the historical trends which may not be

precisely good of the immediate future. In addition, there were some too complicated models that could not be utilized in the interpretation, and this could serve as a barrier to the use of policies.

Siegel et al. (2019)

This paper has expounded on the role of quality housing and social-environmental factors that are related to malaria risk in sub-Saharan Africa that determined that housing is poor as it still continued to play a role in transmission. The researchers analyzed some of these features, which were open eaves, poor walls, the absence of window screens to learn how it contributed to the incidences of malaria infection amongst children based on the data of the Demographic and Health Survey of some countries and logistic regression. The results showed that children living in homes that were in poor constructions were likely to be positive of malaria. The study revealed that housing could be one of the positive interventions to the malaria prevention interventions. However, the cross-sectional design did not permit reaching the provocative conclusion based on the causality, and the additional elaboration in the future investigations is necessary.

Yang et al. (2019)

The study by Yang et al. (2019) explored the correlation that exists between access to safe drinking water and sanitation and malaria risks in children who are less than five years old in Sub-Saharan Africa. The motive behind the identification of this aspect by these authors is that household level of malaria is normally known to be caused by environmental factors hence they are mostly overlooked. The main aim of them was to seek whether there is any connection between poor water and sanitation and the prevalence of malaria, managing socioeconomic variations and climatic fluctuations. The national survey-based study plus logistic regression made

it clear that children who belong to the families that are not able to afford better water or sanitation had higher sets of chances to become malaria-infected. This leads to the significance of water, sanitation and hygiene in reducing the threat of malaria. The analysis, however, failed to add local vectors control data, or spatial clustering of cases of malaria, which explains the relevance of more sophisticated predictive models that will incorporate WASH, vectors ecology, and economic databases.

Research Gaps

Thanks to the analysis of these works, one can say that most of the previous research activity was focused more on the socio-economic and demographic variables. The proposed study will narrow this gap, taking into consideration a larger amount of variables, as well as the treatment, diagnosis, and consultation costs, and the characteristics of women and children, wealth index, and Structured Additive Regression Model. Integration of these factors will be geared toward improving the accuracy of prediction as well as the efforts aimed at assisting to reduce the malaria burden in my family.

2.11 Summary of related literature

Authors (year)	Problem Tackled	Approach Used	Limitation

Tefera et al. (2020)	Estimate malaria burden on households in West Ethiopia	Multivariate logistic regression model using direct and indirect cost data	This study focused only on the socio-economic and socio-demographic features that could contribute to the malaria burden, but did not include other relevant features, such as women's and children's features.
Affiah *et al* (2022)	Estimating Malaria burden on households in Akwa Ibom State	Multivariate logistic regression model using direct and indirect cost data .	The study did not include other features, such as women's and socio-economic features, that could contribute to the malaria burden; it only focused on the cost feature to predict future malaria burdens.

Khan *et al* (2024)	Predicting the various models of malaria	Logistic regression, Random Forest enhance with the support vector machines	The study used secondary data, which may be biased or inaccurate, and the models may not perform well on future outbreaks without regular updates.

Figure 2.1 Summary of related literature